

A First Course in Machine Learning

Chapter 2

Chao-En Yu

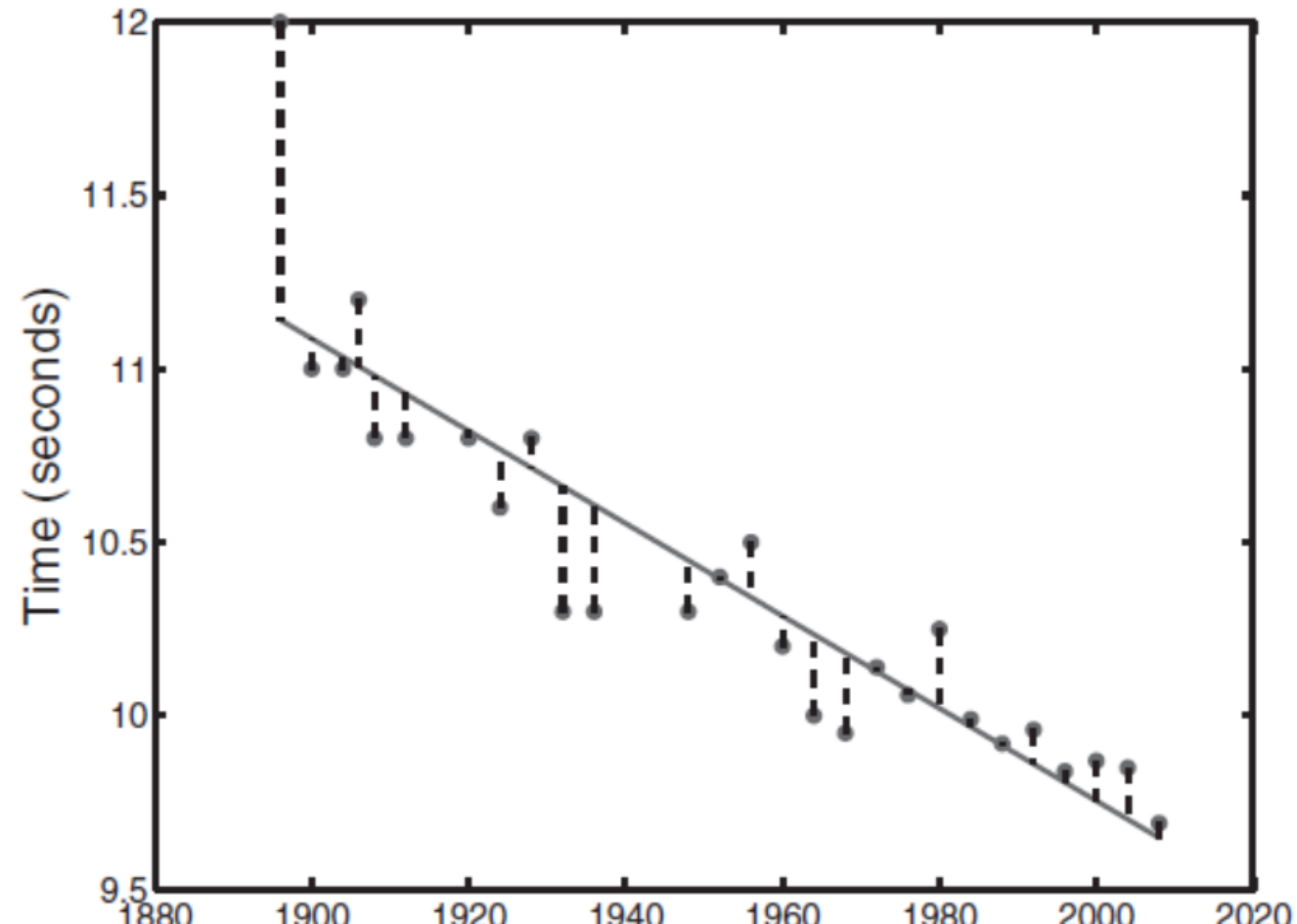
National Tsing Hua University

Drawbacks of OLS

- From a modelling perspective, ignoring the errors between the model and the observations (called noise) is hard to defend. Q: Can you find any ε_i in Ch.1?
- Maximum likelihood approach explicitly models the noise, which allows us to express the level of uncertainty in our estimate of the model parameters $\mathbf{w}$.
- This in turn allows us to express a degree of belief in our predictions, as an interval with a probability.

Generative Model

- Depending on $\mathbf{w}$ and x , our model could be used to generate a winning time for any particular year.



Properties of Errors

1. They are different at each year. Some are positive, some negative and they all have different magnitudes.
2. There does not seem to be any obvious relationship between the size (or direction) of the error and the year.

- We could generate the n th winning time by

$$t_n = \mathbf{w}^T \mathbf{x}_n + \epsilon_n$$

where i.i.d. $\epsilon_n \sim \mathcal{N}(0, \sigma^2 = 0.05)$.

Components of Model

- A deterministic component ($\mathbf{w}^T \mathbf{x}_n$), referred to as a *trend* or a *drift*.

- A random component (ϵ_n), referred to as *noise*:

$$p(\epsilon_1, \dots, \epsilon_N) = \prod_{n=1}^N p(\epsilon_n)$$

$$\epsilon_n \sim \mathcal{N}(0, \sigma^2)$$

Additional parameter

- Addictive Gaussian noise allows us to obtain exact expressions for the optimal parameter value $\hat{\mathbf{w}}$.

Frequentist

- Ronald A. Fisher was the one to combine the ideas of Mendel and Darwin.
- From 1919, he worked at the Rothamsted Experimental Station for 14 years (by designing RCTs and ANOVA).
- In 1922, Fisher introduced the method of maximum likelihood to replace the inverse probability.

And **prior** criteria
in econometrics



The same answers
when prior is
uniform



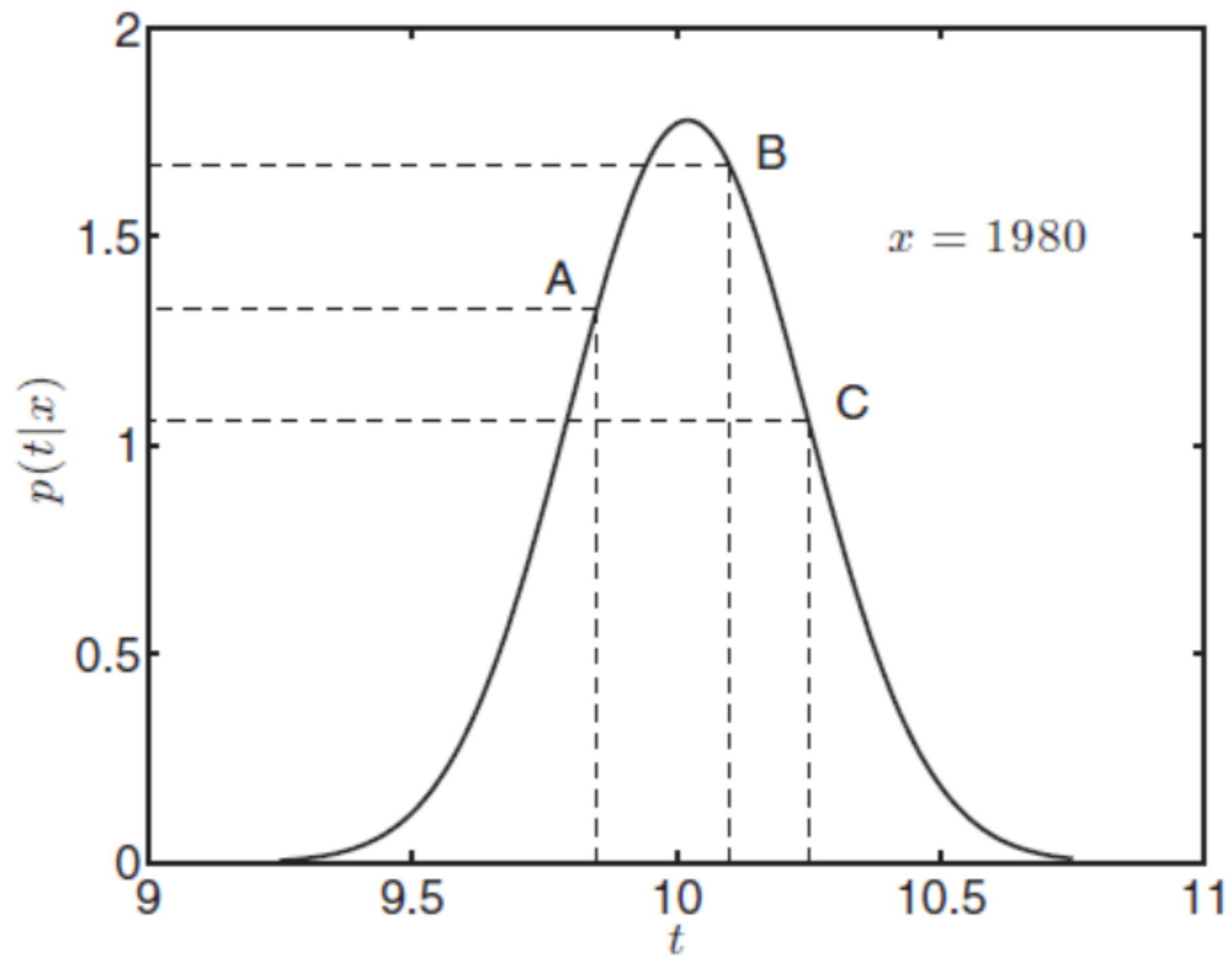
Linear Modelling

- The effect of adding a random variable ϵ_n is that the output of the model, t , is now itself a random variable.
- There is no single value of t_n for a particular $\mathbf{x}_n$, and we cannot use the loss to optimizing $\mathbf{w}$ and σ^2 .
- The continuous random variable t_n has the density

$$p(t_n|\mathbf{x}_n, \mathbf{w}, \sigma^2) = \mathcal{N}(\mathbf{w}^T \mathbf{x}_n, \sigma^2)$$

Is the author
correct here?

→ Note that $p(t_n|\mathbf{x}_n, \mathbf{w}, \sigma^2)$ cannot be interpreted as a probability, though the height indicates how *likely* it is that we would observe a particular t_n for $x = 1980$.



Likelihood

- The density $p(t_n | \mathbf{x}_n, \mathbf{w}, \sigma^2)$ evaluated at t_n is known as the *likelihood* of the n th data point.
 - The most likely winning time in 1980 would be 10.02 seconds. The actual winning time in the 1980 Olympics is C (10.25 seconds).
 - Of A, B and C points, B is the most likely and C the least likely. We cannot change $t_n = 10.25$, so we can only change $\mathbf{w}$ and σ^2 to move the density so as to make it as high as possible at $t_n = 10.25$.
- MLE →

Maximum Likelihood

- If we have N data points, we are interested in the joint conditional density for the whole dataset:

$$p(t_1, \dots, t_N | \mathbf{x}_1, \dots, \mathbf{x}_N, \mathbf{w}, \sigma^2)$$

- Moreover, for the assumption that the noise at each data point is independent, this joint conditional density is a product of N separate terms:

$$L = p(\mathbf{t} | \mathbf{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N p(t_n | \mathbf{x}_n, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(\mathbf{w}^T \mathbf{x}_n, \sigma^2)$$

Independence

- Note that we haven't gone as far as saying that the t_n values are themselves completely independent.
- If they were completely independent, it would not be worthwhile trying to model the data at all.
- The t_n are conditionally independent: Given a value for $\mathbf{w}$, the t_n are independent; without $\mathbf{w}$ they are not independent.

Conditional Independence

- t_{1960} must, in some sense, be dependent on the other data. This dependence is encapsulated in the parameter $\mathbf{w}$.
- If we know $\mathbf{w}$, all that remains is the errors between the observed data and $\mathbf{w}^T \mathbf{x}_n$, which are assumed to be independent.
- Hence, conditioned on $\mathbf{w}$, the observations are independent. Without a model (and therefore a $\mathbf{w}$), the observations are *not* independent.

Maximum Log Likelihood

- A sensible choice of model would be that which maximized the likelihood.
- That is, we will select the model parameters that will *make our observations most likely*. In other words, MLE are *most likely supported by our observations*.
- For analytical reasons, we will maximize the natural logarithm of the likelihood, i.e., $\log L$.

Is the truth in
"our data?"



Maximum Log Likelihood

- Easier to take derivatives →
- After taking logs, we can pull the power down and turn the logarithmic derivative of a product into the sum of the logarithmic derivatives.

$$\begin{aligned}\log L &= \sum_{n=1}^N \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (t_n - f(\mathbf{x}_n; \mathbf{w}))^2 \right\} \right) \\ &= \sum_{n=1}^N \left(-\frac{1}{2} \log 2\pi - \log \sigma - \frac{1}{2\sigma^2} (t_n - f(\mathbf{x}_n; \mathbf{w}))^2 \right) \\ &= -\frac{N}{2} \log 2\pi - N \log \sigma - \frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - f(\mathbf{x}_n; \mathbf{w}))^2\end{aligned}$$

Maximum Likelihood Estimator

- Substituting $f(\mathbf{x}_n|\mathbf{w}) = \mathbf{w}^T \mathbf{x}_n$ gives us the log-likelihood expression as:

$$\log L = -\frac{N}{2} \log 2\pi - N \log \sigma - \frac{1}{2\sigma^2} \sum_{n=1}^N (t_n - \mathbf{w}^T \mathbf{x}_n)^2$$

- Solving the first-order conditions gives us that

$$\hat{\mathbf{w}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{t}$$

is the *maximum likelihood estimator* (MLE) for $\mathbf{w}$
(The OLS maximizes the likelihood when the random error is assumed to follow a normal distribution).

The MLE also →
minimizes the
K-L divergence
for i.i.d. data.